

Vigenere Cipher

Code

```
def vigenere_encrypt(text, key):
    key = key.upper()
    text = text.upper()
    encrypted = ""
    for i in range(len(text)):
        if text[i].isalpha():
            shift = ord(key[i % len(key)]) - 65
            encrypted += chr((ord(text[i]) - 65 + shift) % 26 + 65)
        else:
            encrypted += text[i]
    return encrypted

def vigenere_decrypt(ciphertext, key):
    key = key.upper()
    decrypted = ""
    for i in range(len(ciphertext)):
        if ciphertext[i].isalpha():
            shift = ord(key[i % len(key)]) - 65
            decrypted += chr((ord(ciphertext[i]) - 65 - shift) % 26 + 65)
        else:
            decrypted += ciphertext[i]
    return decrypted

text = input("Enter the Plain Text: ")
key = input("KEY : ")
cipher = vigenere_encrypt(text, key)
print("Encrypted:", cipher)
print("Decrypted:", vigenere_decrypt(cipher, key))
```

```
Vigenere_Cipher.py X AES.py
Vigenere_Cipher.py > ...
13 def vigenere_decrypt(ciphertext, key):
14     decrypted = ""
15     for i in range(len(ciphertext)):
16         if ciphertext[i].isalpha():
17             shift = ord(key[i % len(key)]) - 65
18             decrypted += chr((ord(ciphertext[i]) - 65 - shift) % 26 + 65)
19         else:
20             decrypted += ciphertext[i]
21     return decrypted
22
23
24 text = input("Enter the Plain Text: ")
25 key = input("KEY : ")
26 cipher = vigenere_encrypt(text, key)
27 print("Encrypted:", cipher)
28 print("Decrypted:", vigenere_decrypt(cipher, key))
29

PROBLEMS 20 OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 20

PS C:\Users\thava\OneDrive\Desktop\Currently working\SC> & C:/Users/thava/AppData/Local/Programs/Python/Currently working/SC/Vigenere_Cipher.py
Enter the Plain Text: Thavamani
KEY : Vigenere
Encrypted: OPGZNQRRD
Decrypted: THAVAMANI
PS C:\Users\thava\OneDrive\Desktop\Currently working\SC> |
```